

ARTERN, Germany

WMO: 104600

Lat: 51.3744N

Lon: 11.2919E

Elev: 543

StdP: 14.41

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.6	12.5	3.2	6.6	8.6	8.2	8.5	13.1	31.5	42.1	28.3	40.8	4.3	40	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	84.8	66.8	81.5	65.2	78.4	63.9	68.3	81.0	66.6	78.2	65.1	75.7	8.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.0	91.0	72.7	62.3	85.5	70.7	60.9	81.3	69.8	32.8	81.2	31.4	78.6	30.3	75.7	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	21.5	18.8	6.3	92.5	10.6	3.4	-1.3	95.0	-7.5	97.0	-13.4	98.9	-21.1	101.4		
(5)			5.7	72.0	10.4	2.1	-1.8	73.6	-7.8	74.8	-13.7	76.0	-21.2	77.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	32.6	31.8	40.2	46.6	55.5	60.6	64.9	64.9	58.1	49.5	39.7	35.2	(6)
(7)		DBStd	13.84	10.23	9.66	7.28	6.75	6.22	6.13	5.66	5.74	5.36	6.73	7.05	7.61	(7)
(8)		HDD50	2395	540	511	310	143	22	1	0	0	3	90	314	461	(8)
(9)		HDD65	6246	1004	930	770	552	299	159	74	76	217	480	760	925	(9)
(10)		CDD50	1806	0	1	5	41	191	319	462	461	245	76	4	1	(10)
(11)		CDD65	184	0	0	0	0	3	27	72	72	9	1	0	0	(11)
(12)		CDH74	1854	0	0	0	4	68	274	713	688	101	6	0	0	(12)
(13)		CDH80	459	0	0	0	0	3	58	183	203	12	0	0	0	(13)
(14)	Wind	WSAvg	7.5	9.3	8.7	8.7	8.0	6.8	6.6	6.3	6.2	6.8	7.1	7.7	7.9	(14)
(15)	Precipitation	PrecAvg	22.6	1.5	1.3	1.6	1.4	2.4	2.2	2.8	2.2	2.0	1.5	1.9	1.8	(15)
(16)		PrecMax	31.8	2.7	2.7	3.7	3.5	5.1	4.4	5.4	4.2	5.6	4.7	4.1	3.5	(16)
(17)		PrecMin	14.0	0.2	0.3	0.3	0.3	0.2	0.9	0.7	0.7	0.6	0.3	0.1	0.5	(17)
(18)		PrecStd	3.8	0.6	0.7	0.9	0.7	1.3	0.8	1.2	0.9	1.1	1.0	0.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	58.0	65.0	73.4	79.1	86.0	88.7	91.2	81.9	72.4	57.8	54.0	(19)
(20)			MCWB	47.9	50.5	52.5	56.7	62.3	67.4	69.5	67.4	63.8	61.3	51.9	48.9	(20)
(21)		2%	DB	49.1	51.5	57.5	67.3	75.8	80.9	84.8	85.2	76.8	66.8	54.0	51.2	(21)
(22)			MCWB	45.3	46.2	49.1	54.1	60.6	65.7	67.3	66.2	62.3	58.5	50.1	47.2	(22)
(23)		5%	DB	46.7	47.0	53.5	62.8	72.0	76.9	81.4	81.2	72.1	62.4	51.5	48.6	(23)
(24)			MCWB	43.3	43.2	46.5	51.7	58.9	63.3	65.6	64.8	61.2	55.6	48.1	45.5	(24)
(25)		10%	DB	44.6	43.2	50.4	58.7	68.4	72.7	78.0	77.4	68.1	59.4	49.3	45.8	(25)
(26)			MCWB	41.5	40.0	45.1	49.6	57.2	61.7	64.3	63.5	59.6	53.7	46.2	43.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	50.5	53.1	57.4	64.9	70.5	71.4	70.0	66.8	63.2	53.1	49.6	(27)
(28)			MCDB	52.3	57.3	61.2	70.0	75.9	81.9	85.6	88.3	76.3	71.4	55.8	53.1	(28)
(29)		2%	WB	45.8	46.7	49.9	54.9	62.1	66.8	68.7	67.6	64.2	59.2	50.4	47.6	(29)
(30)			MCDB	48.7	50.7	57.9	65.6	73.0	77.4	81.6	80.7	73.0	65.3	53.3	50.9	(30)
(31)		5%	WB	43.7	43.0	47.4	52.4	59.9	64.8	66.7	66.1	62.2	56.5	48.6	45.8	(31)
(32)			MCDB	46.5	46.6	52.5	61.2	69.9	74.7	78.7	77.7	70.4	61.5	51.5	48.7	(32)
(33)		10%	WB	41.5	40.2	45.3	50.1	57.8	62.5	65.1	64.6	60.1	54.4	46.3	43.2	(33)
(34)			MCDB	44.6	43.0	50.0	58.1	66.9	70.4	75.5	75.3	67.5	58.4	49.0	45.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	10.9	13.3	16.4	18.5	17.2	19.4	19.5	16.9	14.4	9.6	7.4	(35)
(36)			MCDBR	8.9	14.7	20.3	23.3	25.1	24.4	26.2	26.7	24.0	19.9	11.6	9.7	(36)
(37)			MCWBR	7.7	10.9	13.1	12.7	13.1	11.9	11.8	11.5	12.5	12.5	9.0	8.0	(37)
(38)		5% WB	MCDBR	8.8	13.8	19.0	21.2	22.7	22.3	24.0	22.5	21.5	18.9	11.6	9.6	(38)
(39)			MCWBR	8.0	10.5	12.8	12.1	12.5	11.9	11.6	10.5	11.9	12.4	9.2	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.330	0.370	0.408	0.412	0.408	0.424	0.417	0.381	0.354	0.330	0.303	(40)	
(41)		taud	2.406	2.371	2.278	2.215	2.236	2.271	2.231	2.271	2.358	2.424	2.429	2.399	(41)	
(42)		Ebn at Noon	226	251	260	263	267	269	262	257	254	239	214	207	(42)	
(43)		Edh at Noon	19	26	34	41	42	41	42	38	31	24	19	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	233	452	796	1273	1552	1678	1578	1352	978	567	263	176	(44)	
(45)		RadStd	30	62	98	156	180	141	207	131	127	64	35	21	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47)	Regional (49 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	+0.81	N/A	N/A	N/A

Nomenclature: See separate page